

PER2-PROTAC: Data exploration and statistical testing

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1 Purpose

This notebook takes the normalised protein and phosphopeptide abundances produced in notebook 3 and tests whether PROTAC-induced degradation of PER2 changes total protein abundance and, in particular, mTORC1 substrate phosphorylation. It performs statistical testing for both total protein abundance and protein-normalised phosphorylation, then explores the phosphorylation results against reference lists of known mTORC1 substrates to ask whether PER2 degradation preferentially affects mTORC1 substrate phosphosites.

Two ideas recur throughout and are worth stating up front:

- **Very few proteins change in total abundance.** The PROTAC degrades PER2 itself, but does not appreciably change the abundance of most other proteins over the timescale of this experiment; the phosphoproteome is where the more substantial and more numerous changes are seen.

- **Phosphorylation normalised to protein.** A change in a phosphopeptide’s abundance could simply reflect a change in the abundance of its parent protein. To isolate genuine changes in *phosphorylation state*, the phospho test models the protein and phosphopeptide measurements jointly (a **type** factor) and reads off the **condition:type** interaction — the PROTAC effect on phosphorylation *over and above* its effect on the total protein.

Input:

- `notebooks/output/PROTAC_qfs.rds`, `notebooks/output/PROTAC_long_format_protein.rds` and `notebooks/output/PROTAC_long_long_format_phosphopeptide_norm_to_protein.rds` from notebook 3.
- `external/uniprot2details_unique.rds` from notebook 0.
- `raw/PROTAC/mTORC1_substrates_table1_expanded.csv` — a reference table of known mTORC1 substrates and their phosphosites.
- `raw/PROTAC/mTORC1_substrates_Battaglioni2022_plus_2026update.xlsx` — the same substrate reference with additional, more recent substrates added.

Output:

- `results/PROTAC_phospho_limma_results.tsv` (and matching `.rds` in `notebooks/output/`) — the phosphosite-level statistical testing results, for every phosphosite.
- `results/plots/PROTAC_AKT1S1_phosphosite_changes.{png,pdf}` and `..._selected.{png,pdf}` — per-sample phosphorylation change plots for AKT1S1, an mTORC1 substrate with several significantly changing sites.
- `results/plots/PROTAC_AKT1S1_DAP1_phosphosite_changes_selected.{png,pdf}` — a comparison of *in vitro* validated AKT1S1/DAP1 sites against other sites on the same proteins. This filename is produced by two different chunks below (a log2 fold-change version and a linear relative-abundance version); the file on disk reflects whichever was rendered last.
- `results/plots/PROTAC_phospho_limma_results_mTORC1_substrates.{png,pdf}` — the manuscript volcano plot highlighting mTORC1 substrate phosphosites.

Analysis:

1. **Data exploration** — PCA of the protein and phosphopeptide data, to confirm the two conditions separate as expected before proceeding to statistical testing.
2. **Protein statistical testing** — fit `~condition` with `limma`, using `treat()` to additionally require a fold change significantly greater than 1.25x, since we expect only a small number of proteins (predominantly PER2 itself) to change substantially in abundance.
3. **Phosphopeptide statistical testing** — model protein and phosphopeptide measurements jointly with `~condition*type` (blocking on the paired protein/phosphopeptide measurements from the same sample) and test the **condition:type** interaction, i.e. the PROTAC effect on phosphorylation after accounting for any change in protein abundance.
4. **Result summaries** — the overall number and direction of significant phosphosite changes, a volcano plot, and detailed per-site plots for AKT1S1 (and, for validation purposes, DAP1), two proteins with multiple significantly changing, well-characterised mTORC1-regulated phosphosites.
5. **mTORC1 substrate enrichment** — an exploratory, then a more refined, comparison of phosphosite fold changes against reference lists of known mTORC1 substrates, testing whether substrate phosphosites are disproportionately represented among the significant changes.
6. **Export** — write the phosphosite-level statistics table to `results/`.

How to read the results: significance for the primary phosphosite testing is assessed at 5% FDR ($\text{adj.P.Val} < 0.05$) in the early result-summary sections, but the mTORC1 substrate enrichment sections (from the volcano plot with mTORC1 substrates highlighted onwards) use a more permissive 10%

FDR threshold, reflecting the smaller number of substrate phosphosites available to test against and the exploratory nature of that analysis. Positive log2 fold-changes indicate higher phosphorylation (or protein abundance) in PROTAC relative to DMSO.

2 Setup

Load required packages

Define directory for plots

```
plots_dir <- here('results/plots')
if(!dir.exists(plots_dir)){
  dir.create(plots_dir, recursive = TRUE)
}
```

Read in the data

```
output_dir <- here('notebooks/output')

qfs <- readRDS(here(output_dir, 'PROTAC_qfs.rds'))

long_format_protein <- readRDS(
  here(output_dir, 'PROTAC_long_format_protein.rds'))

long_long_format_phosphopeptide_norm_to_protein <- readRDS(
  here(output_dir, 'PROTAC_long_long_format_phosphopeptide_norm_to_protein.rds'))

uniprot2details_unique <- readRDS(here('external/uniprot2details_unique.rds'))
```

3 Data exploration

3.1 PCA plots

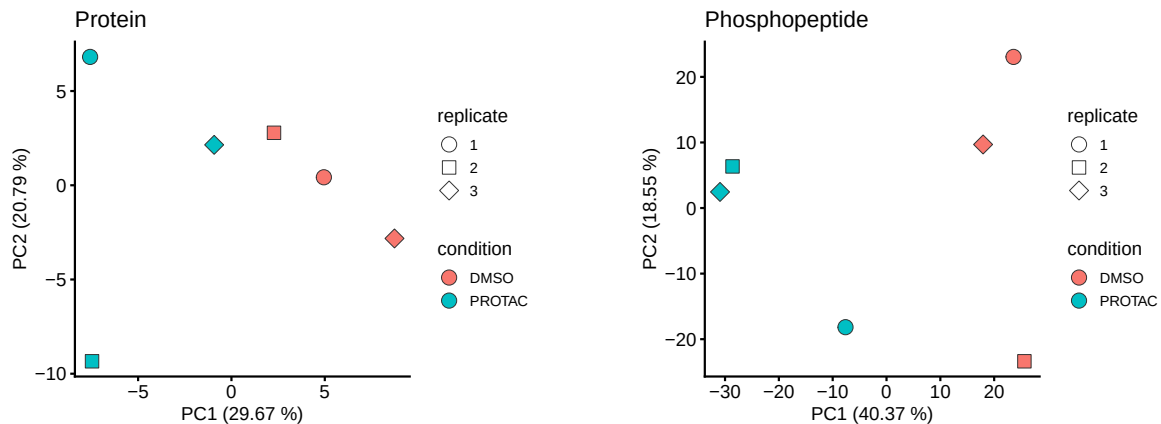
Below, we make PCA plots for the protein and phosphopeptide data. At the protein level, DMSO and PROTAC separate on PC1 (30% of variance), with two DMSO-2 and PROTAC-3 relatively close together. For phosphopeptides, the conditions separate very well (PC1; ~40% variance explained)

```
for(qf in names(qfs)){

  if(qf=='total'){
    assay_name <- 'protein'
    feature <- 'Protein'
  } else{
    assay_name <- 'phosphopeptide'
    feature <- 'Phosphopeptide'
  }

  print(plot_pca(
    qfs[[qf]], i=assay_name, colour_by = 'condition', shape='replicate') +
```

```
ggtitle(feature))
}
```



4 Statistical testing of protein abundance changes between conditions

4.1 Protein

First, we'll perform statistical testing to compare the conditions (PROTAC vs DMSO) for total protein abundance. Alongside the standard `eBayes` moderation, we also fit a `treat()` model, which tests against a minimum fold-change threshold (here, 1.25x) rather than a null of exactly zero change. This is appropriate here since we are not primarily interested in detecting arbitrarily small but statistically significant changes in abundance across the whole proteome, only in changes large enough to plausibly reflect a genuine biological consequence of PER2 degradation; `treat()` is applied to the `lmFit` output directly, without a prior `eBayes` call.

```
## Create design matrix for the linear model
condition <- factor(qfs$total[['protein']]$condition, levels=c('DMSO', 'PROTAC'))
m_design <- model.matrix(~condition)

## Fit linear model using the design matrix and desired contrasts
fit_model <- lmFit(object = assay(qfs$total[['protein']] ), design = m_design)

## Update the model using the limma eBayes algorithm
final_model <- eBayes(fit = fit_model,
                      trend = TRUE,
                      robust = TRUE)

## Add TREAT to final model to perform fold-change thresholding for contrasts
# require |FC| significantly over 1.25x
final_model_treat <- treat(fit_model, fc=1.25, trend=TRUE, robust=TRUE)

## QC plot to assess if there is a trend between abundance and variance, as expected
plotSA(fit = final_model,
       cex = 0.5,
```

```

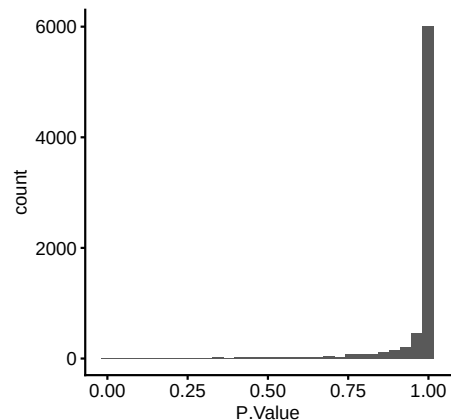
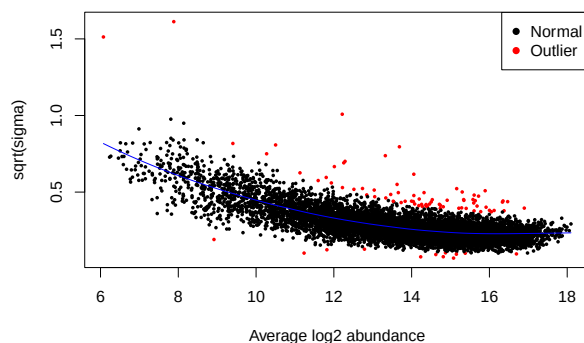
xlab = "Average log2 abundance")

## Format results
prot_limma_results <- topTreat(
  fit = final_model_treat,
  coef = 'conditionPROTAC',
  adjust.method = "BH", # Method for multiple hypothesis testing
  number = Inf) %>% # Print results for all proteins
  tibble::rownames_to_column("Protein") %>%
  merge(data.frame(rowData(qfs$total)[['protein']]),
        by.x='Protein', by.y='row.names', all.x=TRUE) %>%
  arrange(P.Value)

## Check distribution of p-values. Because we are using treat, we don't expect a
## uniform distribution under the null, but rather a peak near 1.
p <- prot_limma_results %>%
  as_tibble() %>%
  ggplot(aes(x = P.Value)) +
  geom_histogram() +
  theme_biomasslmb()

print(p)

```



How many proteins have a significant change in abundance

```

table(prot_limma_results$adj.P.Val<0.05,
      ifelse(prot_limma_results$logFC>0, 'Increased', 'Decreased')) %>%
  knitr::kable()

```

	Decreased	Increased
FALSE	3586	4060
TRUE	1	1

```

prot_limma_results %>%
  filter(adj.P.Val<0.05) %>%
  select(Protein, Gene.Names.First, logFC, adj.P.Val) %>%
  knitr::kable()

```

Protein	Gene.Names.First	logFC	adj.P.Val
O00469	PLOD2	-0.8884707	7.00e-07
Q16850	CYP51A1	0.7847998	4.25e-05

Very few proteins have a significant change in abundance.

4.2 Phosphopeptide statistical testing

Next, we perform statistical testing for the phosphopeptide data, comparing PROTAC vs DMSO while accounting for the matched protein abundance measurements. The data matrix below has one column per condition/replicate/type combination (`type` being `protein` or `phosphopeptide`), so each phosphosite's row includes both its own phosphopeptide abundance and the abundance of its parent protein for the same samples. Fitting `~condition * type` and testing the `condition:type` interaction coefficient asks whether the PROTAC effect differs between the phosphopeptide and protein measurements — i.e. whether phosphorylation changes beyond what is explained by any change in protein abundance. Since the protein and phosphopeptide measurements for a given sample are not independent (they derive from the same biological replicate), we block on `condition:replicate` and estimate the within-block correlation with `duplicateCorrelation` before fitting.

```
# Create data matrix for modelling
to_model_phospho <- long_long_format_phosphopeptide_norm_to_protein %>%
  filter(type!='phosphopeptide_prot_norm') %>%
  select(prot_phos, condition, replicate, type, abundance) %>%
  pivot_wider(names_from = c(condition, replicate, type), values_from=abundance) %>%
  tibble::column_to_rownames('prot_phos')

# Create design matrix for the linear model
exp_cond_phospho <- data.frame(sample=colnames(to_model_phospho)) %>%
  separate(sample, into=c('condition', 'replicate', 'type'), remove = FALSE)

condition <- factor(exp_cond_phospho$condition, levels=c('DMSO', 'PROTAC'))

type <- factor(exp_cond_phospho$type, levels=c('protein', 'phosphopeptide'))

m_design_phospho <- model.matrix(~ condition * type)

# same ID for phospho + total from the same samples
block <- interaction(condition, exp_cond_phospho$replicate)

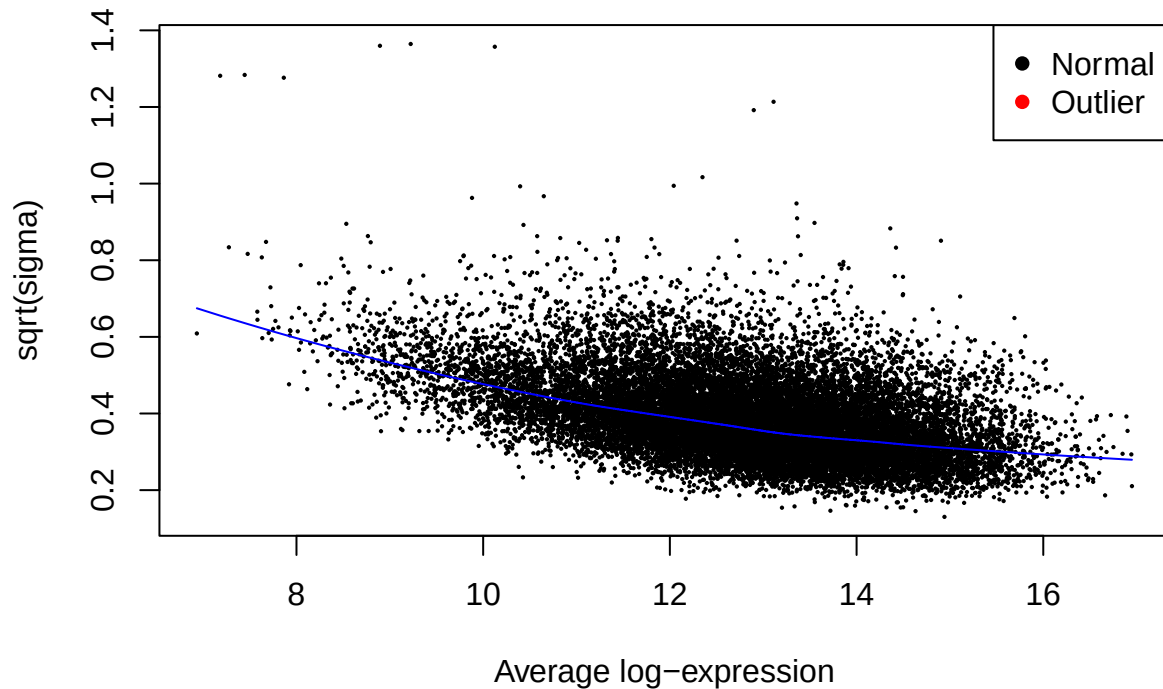
# Estimate the correlation between measurements from the same block
corfit <- duplicateCorrelation(as.matrix(to_model_phospho),
                              m_design_phospho, block=block)

# Fit the linear model with blocking and correlation information
limma_fit_phospho <- lmFit(
  as.matrix(to_model_phospho), m_design_phospho, block=block, correlation=corfit$consensus)

# Update the model using the limma eBayes algorithm
limma_fit_phospho_eb <- eBayes(limma_fit_phospho, trend=TRUE, robust=TRUE)

# QC plot to assess if there is a trend between abundance and variance, as expected,
```

```
# and if the limma 'trend' captures this
plotSA(limma_fit_phospho_eb)
```



```
# Format results
phospho_limma_results <- topTable(limma_fit_phospho_eb, coef = 'conditionPROTAC:typephosphopeptide',
  adjust.method = "BH",      # Method for multiple hypothesis testing
  number = Inf) %>%         # Print results for all proteins
  tibble::rownames_to_column("phosphosite") %>%
  merge(rowData(qfs$phospho)[['phosphopeptide']][
    ,c('Entry.Name', 'Protein.names.y', 'Gene.Names', 'Gene.Names.First', 'site_seq')],
    by.x='phosphosite', by.y='row.names', all.x=TRUE) %>%
  data.frame() %>%
  separate(phosphosite, into=c('UniprotID', 'phospho_pos'), sep='_', remove=FALSE)
```

Save results

```
saveRDS(phospho_limma_results, here(output_dir, 'PROTAC_phospho_limma_results.rds'))
```

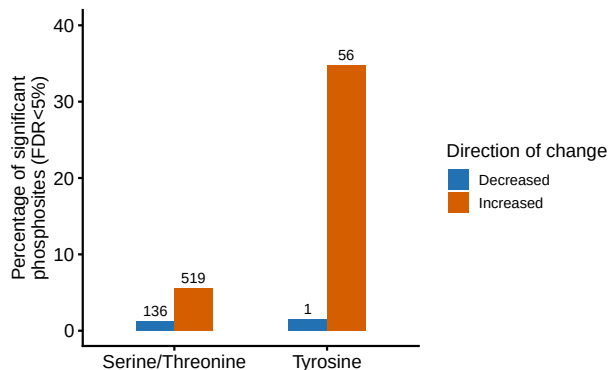
4.3 How many phosphosites have a significant change in abundance?

Below, we summarise how many phosphosites have a significant change in abundance (FDR<5%), and how the balance of increased vs decreased sites differs between serine/threonine and tyrosine phosphorylation.

```
table(ifelse(phospho_limma_results$adj.P.Val<0.05, 'Significant', 'Not significant'),
      ifelse(phospho_limma_results$logFC>0, 'Increased', 'Decreased')) %>%
  knitr::kable()
```

	Decreased	Increased
Not significant	10377	9007
Significant	137	575

```
phospho_limma_results %>%
  mutate(type=ifelse(grepl('Y', phospho_pos), 'Tyrosine', 'Serine/Threonine')) %>%
  mutate(sig=adj.P.Val<0.05) %>%
  group_by(sig, type, direction=ifelse(logFC>0, 'Increased', 'Decreased')) %>%
  tally() %>%
  group_by(type, direction) %>%
  mutate(perc=100*n/sum(n)) %>%
  filter(sig) %>%
  ungroup() %>%
  complete(direction, type, fill=list(n=0)) %>%
  ggplot(aes(type, perc, fill=direction)) +
  geom_col(position='dodge', width=0.5) +
  theme_biomasslmb() +
  labs(x='', y='Percentage of significant phosphosites (FDR<5%)', fill='Direction of change') +
  scale_fill_manual(values=get_cat_palette(2)) +
  geom_text(aes(label=n), position=position_dodge(width=0.5), vjust=-0.5, size=4) +
  ylim(0, 40) +
  theme(strip.background = element_blank())
```



4.4 Volcano plot

Below, we make a volcano plot for the statistical test results, labelling the 10 most significant phosphosites in each direction of change.

```
# highlight top 10 significant proteins per genotype
phospho_limma_results %>%
  group_by(logFC>0) %>%
```

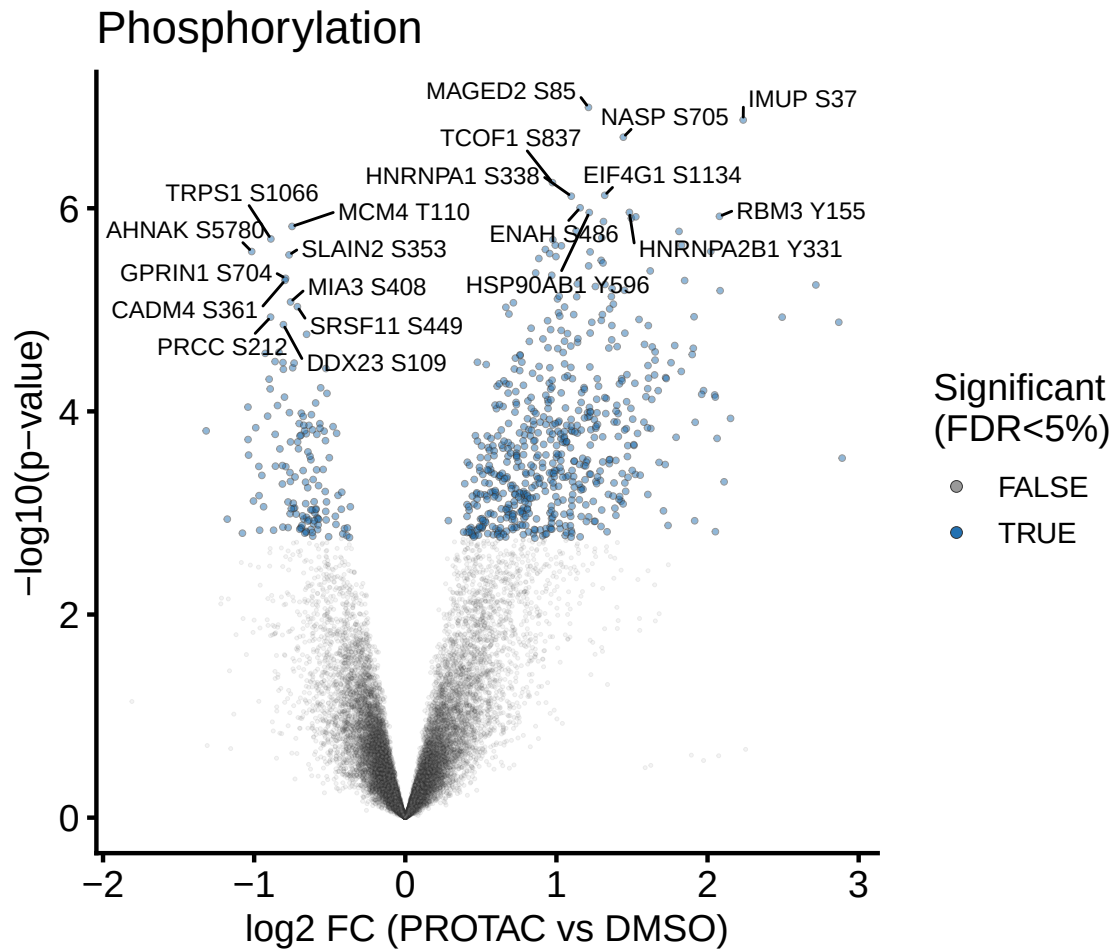
```

filter(adj.P.Val<0.05) %>%
slice_min(P.Value, n=10) -> highlight

phospho_limma_results %>%
mutate(significant=adj.P.Val<0.05) %>%
ggplot(aes(logFC, -log10(P.Value))) +
geom_point(aes(fill=significant,
               alpha=significant, size=significant),
           colour='grey20', stroke=0.25, shape=21) +
theme_biomasslmb() +
scale_fill_manual(values=c('grey60', get_cat_palette(1)),
                  name='Significant\n(FDR<5%)') +
scale_size_manual(values=c(0.5, 1), guide='none') +
scale_alpha_manual(values=c(0.1, 0.5), guide='none') +
labs(x='log2 FC (PROTAC vs DMSO)', y='-log10(p-value)') +
theme(strip.background = element_blank()) +
geom_text_repel(
  data=highlight,
  aes(label=paste(Gene.Names.First, gsub('.*_', '', phosphosite))),
  max.overlaps = 50, size=3, min.segment.length =0, colour='black', force = 10) +
ggtitle('Phosphorylation') -> p

print(p)

```



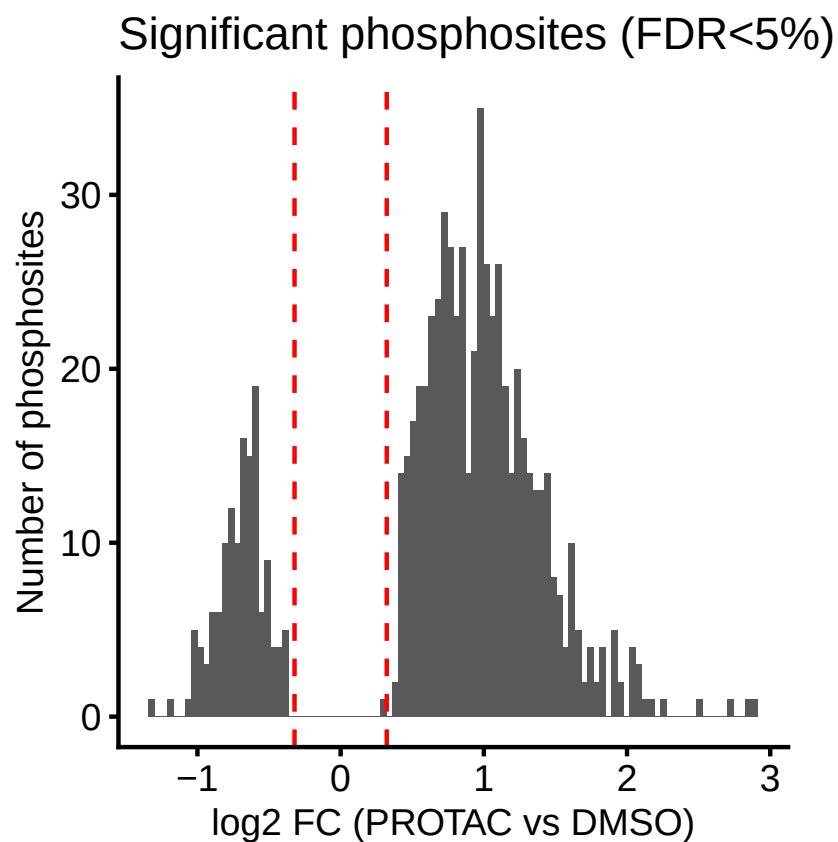
The plots below examine the magnitude of significant changes in more detail: a crosstabulation against a 1.1-fold change threshold, and the distribution of fold changes among significant phosphosites relative to the 1.25-fold change threshold used elsewhere in this notebook.

```
table(dir=ifelse(phospho_limma_results$logFC>0, 'inc', 'dec'), sig=phospho_limma_results$adj.P.Val<0.05
```

```
## , , over_1.1 = FALSE
##
##      sig
## dir  FALSE TRUE
## dec  5427    0
## inc  4321    0
##
## , , over_1.1 = TRUE
##
##      sig
## dir  FALSE TRUE
```

```
## dec 4950 137
## inc 4686 575
```

```
phospho_limma_results %>%
  filter(adj.P.Val<0.05) %>%
  ggplot(aes(logFC)) +
  geom_histogram(bins=100) +
  theme_biomasslmb() +
  labs(x='log2 FC (PROTAC vs DMSO)', y='Number of phosphosites') +
  theme(strip.background = element_blank()) +
  ggtitle('Significant phosphosites (FDR<5%)') +
  geom_vline(xintercept = log2(1.25), colour='red', linetype='dashed') +
  geom_vline(xintercept = -log2(1.25), colour='red', linetype='dashed')
```



4.5 Plot changes in phosphorylation for AKT1S1

Below, we plot the changes in phosphorylation for AKT1S1, which has multiple significantly changing sites and is itself a well-characterised direct mTORC1 substrate. Values are expressed relative to the mean DMSO abundance for each site (`abundance_diff`), restricted to sites significant at 10% FDR.

```
p_values <- phospho_limma_results %>%
  filter(Gene.Names.First=='AKT1S1') %>%
  mutate(ptm_name=phosphosite)
```

```

to_plot <- long_long_format_phosphopeptide_norm_to_protein %>%
  filter(Gene.Names.First=='AKT1S1', type=='phosphopeptide_prot_norm')

DMSO_abundances <- to_plot %>%
  filter(condition=='DMSO') %>%
  group_by(prot_phos) %>%
  summarise(abundance=mean(abundance))

to_plot <- to_plot %>%
  merge(DMSO_abundances, by='prot_phos', suffix=c('', '_DMSO')) %>%
  mutate(abundance_diff=abundance - abundance_DMSO) %>%
  group_by(condition, prot_phos) %>%
  mutate(mean_abundance_diff=mean(abundance_diff)) %>%
  arrange(desc(mean_abundance_diff)) %>%
  mutate(ptm_name_facet=gsub(';', ' ', '\n', ptm_name))

to_plot$ptm_name_facet <- factor(to_plot$ptm_name_facet,
                                levels=unique(to_plot$ptm_name_facet))

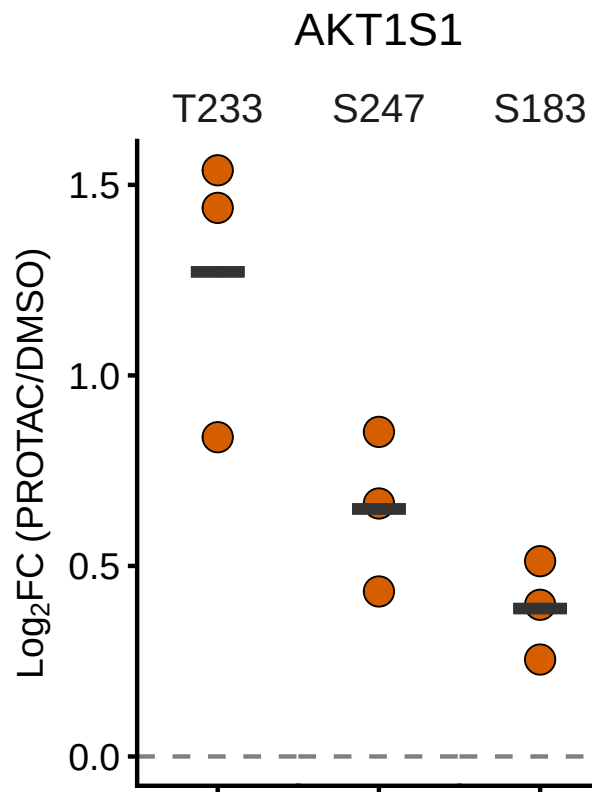
p_values <- phospho_limma_results %>%
  filter(Gene.Names.First=='AKT1S1')

sig_sites <- p_values %>%
  filter(adj.P.Val<0.1) %>%
  pull(phospho_pos)

p <- to_plot %>%
  filter(ptm_name %in% sig_sites) %>%
  filter(condition!='DMSO') %>%
  ggplot(aes(ptm_name, abundance_diff)) +
  geom_hline(yintercept = 0, colour='grey50', linetype='dashed') +
  geom_point(colour='black', stroke=0.5, shape=21,
             fill=get_cat_palette(2)[2], size=5) +
  stat_summary(fun.min=mean, fun.max=mean, geom='errorbar',
              linewidth=2, width=0.4, colour='grey20') +
  theme_biomasslmb(aspect_square = FALSE) +
  labs(x='', y=expression(Log[2]*FC~"(PROTAC/DMSO)")) +
  theme(axis.text.x=element_text(angle=45, hjust=1, vjust=1)) +
  facet_grid(.~ptm_name_facet, scales='free_x') +
  theme(strip.background = element_blank(),
        panel.spacing = unit(0, 'mm'),
        axis.text.x = element_blank(),
        aspect.ratio=4,
        strip.text.x=element_text(vjust=1, hjust=0),
        plot.title = element_text(hjust=0.5)) +
  ggtitle('AKT1S1')

p1 <- p + theme(strip.text.x=element_text(vjust=0.5, hjust=0.5, size=15))
print(p1)

```



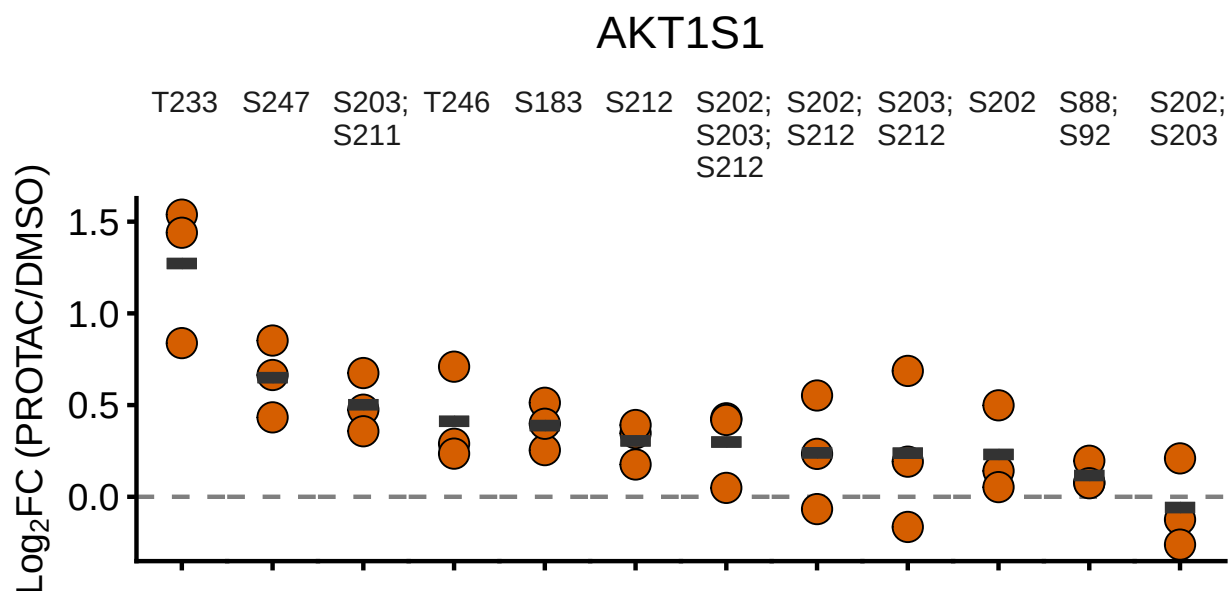
```
ggsave(plot=p1, here('results/plots/PROTAC_AKT1S1_phosphosite_changes_selected.png'))
```

```
## Saving 6.5 x 4.5 in image
```

```
ggsave(plot=p1, here('results/plots/PROTAC_AKT1S1_phosphosite_changes_selected.pdf'))
```

```
## Saving 6.5 x 4.5 in image
```

```
p2 <- p + filter(to_plot, condition!='DMSO')
print(p2)
```



```
ggsave(plot=p2, here('results/plots/PROTAC_AKT1S1_phosphosite_changes.png'))
```

```
## Saving 6.5 x 4.5 in image
```

```
ggsave(plot=p2, here('results/plots/PROTAC_AKT1S1_phosphosite_changes.pdf'))
```

```
## Saving 6.5 x 4.5 in image
```

4.6 Selected validated phosphosites: AKT1S1 and DAP1

The next three chunks focus on a small set of phosphosites with prior *in vitro* validation as direct mTORC1 substrates: two DAP1 sites (P51397_S2;S3, P51397_S49;S51) and AKT1S1 S183 (Q96B36_S183), alongside nearby sites on the same proteins for comparison. The first chunk below compares the validated sites (S183, T233, S247 on AKT1S1, and both DAP1 sites) against other sites on the same two proteins, as log₂ fold changes. The following two chunks re-plot a restricted set of the same sites (dropping T233/S247) on an untransformed, DMSO-relative abundance scale — first as fold-change-from-DMSO, then as raw protein-normalised abundance per sample — as alternative visualisations of the same underlying values. All three chunks save to the same `PROTAC_AKT1S1_DAP1_phosphosite_changes_selected` filename, so the file on disk reflects whichever of these was rendered most recently (the untransformed relative-abundance version, from the second of the three chunks).

```
phoi <- c('P51397_S2;S3', 'P51397_S49;S51', 'Q96B36_S183')
```

```

to_plot <- long_long_format_phosphopeptide_norm_to_protein %>%
  filter(phosphopeptide %in% phoi, type=='phosphopeptide_prot_norm')

DMSO_abundances <- to_plot %>%
  filter(condition=='DMSO') %>%
  group_by(prot_phos) %>%
  summarise(abundance=mean(abundance))

to_plot <- to_plot %>%
  merge(uniprot2details_unique[c('UniprotID', 'Entry.Name')], by.x='UniprotID', by.y='UniprotID') %>%
  merge(DMSO_abundances, by='prot_phos', suffix=c('', '_DMSO')) %>%
  mutate(abundance_diff=abundance - abundance_DMSO) %>%
  group_by(condition, prot_phos) %>%
  mutate(mean_abundance_diff=mean(abundance_diff)) %>%
  arrange(desc(mean_abundance_diff)) %>%
  mutate(ptm_name_facet=sprintf('%s\n%s', sub('_HUMAN', '', Entry.Name), sub('; ', ';', ptm_name)))

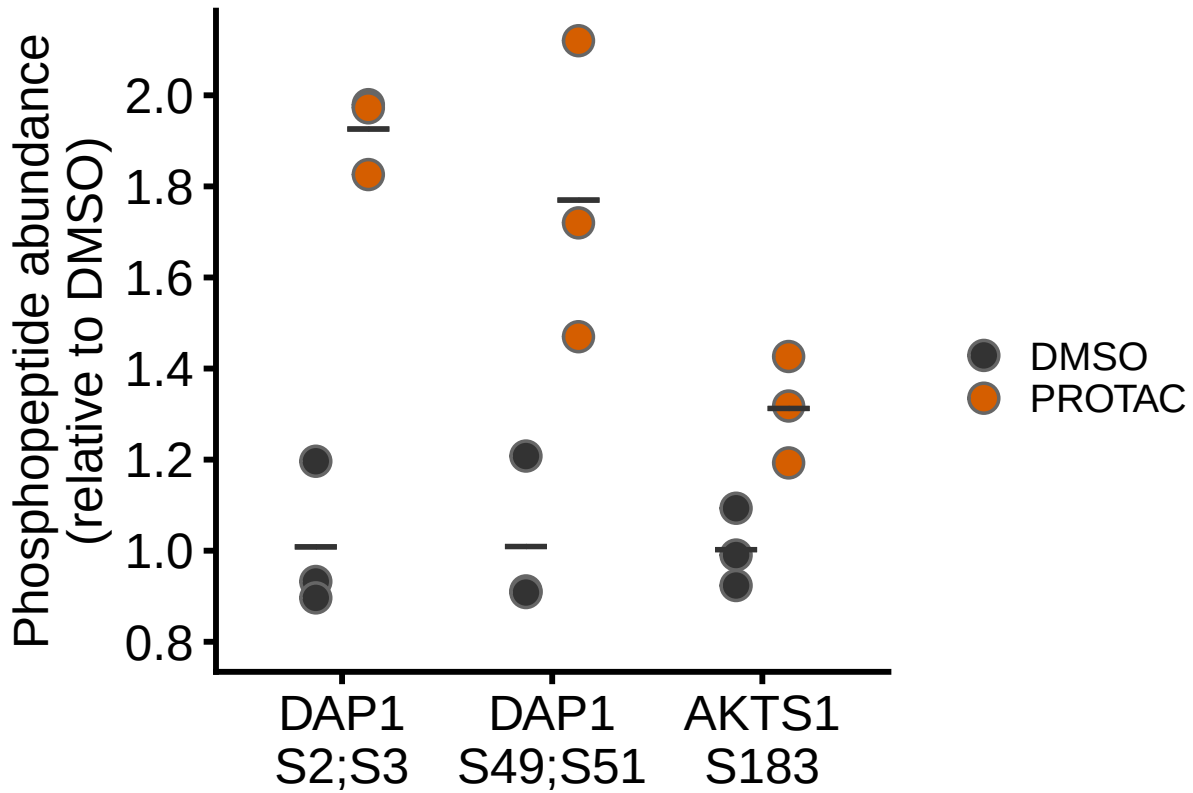
to_plot$ptm_name_facet <- factor(to_plot$ptm_name_facet,
                                levels=unique(to_plot$ptm_name_facet))

val_sites <- phoi[1:3]

p <- to_plot %>%
  #filter(condition!='DMSO') %>%
  ggplot(aes(ptm_name_facet, 2^abundance_diff, fill=condition, group=interaction(condition, ptm_name_facet)),
    geom_point(stroke=1, shape=21, size=5, position=position_dodge(width=0.5), colour='grey40') +
    scale_fill_manual(values=c('grey20', get_cat_palette(2)[2]), name='') +
    stat_summary(fun.min=mean, fun.max=mean, geom='errorbar',
      linewidth=1, width=0.4, colour='grey20',
      position=position_dodge(width=0.5)) +
    theme_biomasslmb(aspect_square = FALSE, base_size = 20) +
    labs(x='', y="Phosphopeptide abundance\n(relative to DMSO)") +
    theme(strip.background = element_blank(),
      strip.text = element_blank(),
      panel.spacing = unit(10, 'mm'),
      strip.text.x=element_text(vjust=1, hjust=0),
      plot.title = element_text(hjust=0.5)) +
    theme(strip.text.x=element_blank()) + scale_y_continuous(limits=c(0.8, NA), breaks=seq(0.8, 2.2, 0.2))

print(p)

```



```
ggsave(plot=p, here('results/plots/PROTAC_AKT1S1_DAP1_phosphosite_changes_selected.png'))
```

```
## Saving 7 x 5 in image
```

```
ggsave(plot=p, here('results/plots/PROTAC_AKT1S1_DAP1_phosphosite_changes_selected.pdf'))
```

```
## Saving 7 x 5 in image
```

5 mTORC1 substrate enrichment: manuscript figure

This section repeats the substrate-enrichment analysis using the updated substrate reference table (`mTORC1_substrates_Battaglioni2022_plus_2026update.xlsx`), and distinguishes three categories of phosphosite for the final manuscript figure: sites with direct *in vitro* validation as mTORC1 substrates (`highlight_val`, matched against the updated table, plus the two DAP1 sites already highlighted above), other sites on the same validated proteins from the original substrate table (`highlight_all_sites_in_protein`), and everything else. DAP1 (P51397, gene name `DAP`) and the AKT1S1 (Q96B36) sites are labelled explicitly on the resulting plot, which is saved as the manuscript volcano figure. `set.seed(0)` is included for reproducibility, although no random sampling is performed in this chunk.

```

set.seed(0)

mtor_substrates <- read.csv(here('raw/PROTAC/mTORC1_substrates_table1_expanded.csv'))

mtor_substrates_updated <- openxlsx::read.xlsx(here('raw/PROTAC/mTORC1_substrates_Battaglioni2022_plus_1.xlsx'),
  data.frame())

nrow(mtor_substrates)

## [1] 107

nrow(mtor_substrates_updated)

## [1] 136

merge(mtor_substrates, mtor_substrates_updated, by=c('Uniprot.ID', 'Phosphorylated.site.s.')) %>%
  nrow()

## [1] 107

phospho_limma_results %>%
  merge(mtor_substrates_updated,
    by.x=c('UniprotID', 'phospho_pos'),
    by.y=c('Uniprot.ID', 'Phosphorylated.site.s.')) %>%
  select(Gene.Names.First, Entry.Name)

##      Gene.Names.First  Entry.Name
## 1          PRKAB2  AAKB2_HUMAN
## 2          TBC1D4  TBCD4_HUMAN
## 3              AR   ANDR_HUMAN
## 4          EIF2S2  IF2B_HUMAN
## 5          STAT3  STAT3_HUMAN
## 6          PRKAA2  AAPK2_HUMAN
## 7          TRIM28  TIF1B_HUMAN
## 8      EIF4EBP1  4EBP1_HUMAN
## 9      EIF4EBP1  4EBP1_HUMAN
## 10     EIF4EBP1  4EBP1_HUMAN
## 11     EIF4EBP1  4EBP1_HUMAN
## 12          JMJD1C  JHD2C_HUMAN
## 13          RPTOR  RPTOR_HUMAN
## 14          RPTOR  RPTOR_HUMAN
## 15          SERBP1  SERB1_HUMAN
## 16          AKT1S1  AKTS1_HUMAN
## 17          AKT1S1  AKTS1_HUMAN
## 18          SIRT1  SIR1_HUMAN
## 19          MAF1  MAF1_HUMAN
## 20          MAF1  MAF1_HUMAN
## 21          ISCU  ISCU_HUMAN
## 22          UVRAG  UVRAG_HUMAN
## 23          ACIN1  ACINU_HUMAN
## 24          ACIN1  ACINU_HUMAN

```

```
## 25          USP20  UBP20_HUMAN
## 26          IGF2BP2 IF2B2_HUMAN
## 27          IGF2BP2 IF2B2_HUMAN
```

```
# highlight
highlight_val <- phospho_limma_results %>%
  merge(mtor_substrates_updated,
        by.x=c('UniprotID', 'phospho_pos'),
        by.y=c('Uniprot.ID', 'Phosphorylated.site.s.')) %>%
  select(logFC, P.Value, adj.P.Val, Gene.Names.First, Entry.Name, phosphosite) %>%
  mutate(type='in vitro validated')

highlight_val2 <- phospho_limma_results %>%
  filter(phosphosite %in% c('P51397_S2;S3', 'P51397_S49;S51')) %>%
  select(logFC, P.Value, adj.P.Val, Gene.Names.First, Entry.Name, phosphosite) %>%
  mutate(type='in vitro validated')

highlight_val <- bind_rows(highlight_val, highlight_val2) %>%
  distinct()

highlight_all_sites_in_protein <- phospho_limma_results %>%
  filter(UniprotID %in% unique(mtor_substrates$Uniprot.ID)) %>%
  filter(!phosphosite %in% highlight_val$phosphosite) %>%
  select(logFC, P.Value, adj.P.Val, Gene.Names.First, Entry.Name, phosphosite) %>%
  mutate(type='other site on validated protein')

highlight <- bind_rows(highlight_val, highlight_all_sites_in_protein) %>%
  mutate(significant=adj.P.Val<0.1)

highlight %>%
  filter(Gene.Names.First=='AKT1S1')
```

##	logFC	P.Value	adj.P.Val	Gene.Names.First	Entry.Name
## 1	0.38828008	0.0039193037	0.08032776	AKT1S1	AKTS1_HUMAN
## 2	0.30470177	0.0174740857	0.19311543	AKT1S1	AKTS1_HUMAN
## 3	0.23077510	0.2762929065	0.68868626	AKT1S1	AKTS1_HUMAN
## 4	-0.05863947	0.7837159438	0.94237166	AKT1S1	AKTS1_HUMAN
## 5	0.29842627	0.0753778107	0.41417673	AKT1S1	AKTS1_HUMAN
## 6	0.23970113	0.2884430766	0.69895130	AKT1S1	AKTS1_HUMAN
## 7	0.50224098	0.1410958315	0.53763023	AKT1S1	AKTS1_HUMAN
## 8	0.23772754	0.3082356346	0.71544751	AKT1S1	AKTS1_HUMAN
## 9	0.64961021	0.0016238829	0.04709026	AKT1S1	AKTS1_HUMAN
## 10	0.11611963	0.4860088632	0.82129449	AKT1S1	AKTS1_HUMAN
## 11	1.27191851	0.0004413551	0.02088897	AKT1S1	AKTS1_HUMAN
## 12	0.41125695	0.0180523901	0.19535855	AKT1S1	AKTS1_HUMAN

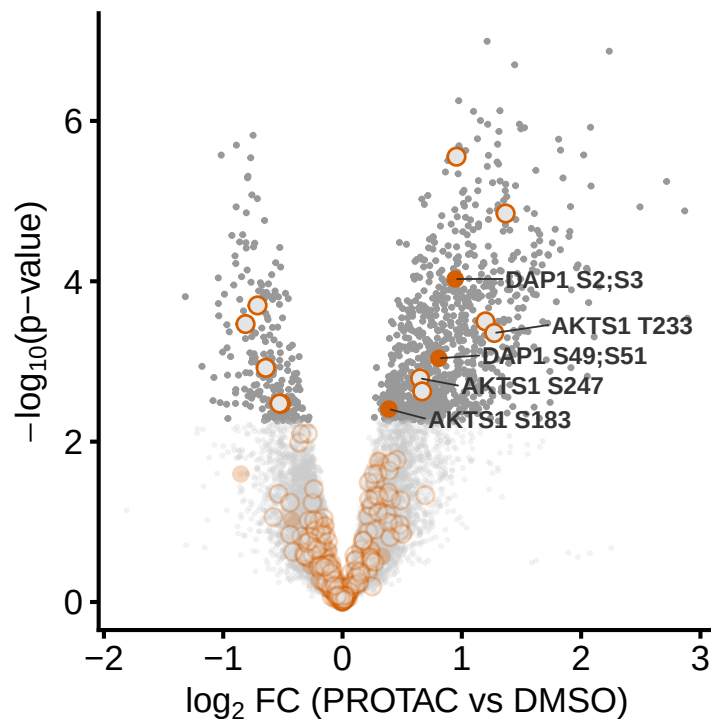
##	phosphosite	type	significant
## 1	Q96B36_S183	in vitro validated	TRUE
## 2	Q96B36_S212	in vitro validated	FALSE
## 3	Q96B36_S202	other site on validated protein	FALSE
## 4	Q96B36_S202;S203	other site on validated protein	FALSE
## 5	Q96B36_S202;S203;S212	other site on validated protein	FALSE
## 6	Q96B36_S202;S212	other site on validated protein	FALSE
## 7	Q96B36_S203;S211	other site on validated protein	FALSE

```
## 8      Q96B36_S203;S212 other site on validated protein      FALSE
## 9      Q96B36_S247 other site on validated protein          TRUE
## 10     Q96B36_S88;S92 other site on validated protein      FALSE
## 11     Q96B36_T233 other site on validated protein          TRUE
## 12     Q96B36_T246 other site on validated protein          FALSE
```

```
phospho_limma_results %>%
  filter(adj.P.Val<0.1) %>%
  ggplot(aes(logFC, -log10(P.Value))) +
  geom_point(colour='grey60', alpha=1, size=1, shape=20) +
  geom_point(data = filter(phospho_limma_results, adj.P.Val>0.1),
             colour='grey80', alpha=0.25, size=0.5, shape=20) +

  geom_point(data=filter(highlight, type=='in vitro validated'), colour=get_cat_palette(2)[2],
             aes(alpha=significant), pch=16, size=4) +
  geom_point(data=filter(highlight, type=='other site on validated protein'),
             colour=get_cat_palette(2)[2],
             fill='grey90',
             aes(alpha=significant), pch=21, size=4) +
  theme_biomasslmb(base_size = 20) +
  scale_fill_manual(values=c(get_cat_palette(2)[2], 'grey90'), name='mTORC1 substrate') +
  scale_size_manual(values=c(0.5, 1), guide='none') +
  scale_alpha_manual(values=c(0.25, 1), guide='none') +
  labs(x = expression(log[2]*" FC (PROTAC vs DMSO)"),
       y = expression(-log[10]*"(p-value)")) +
  theme(strip.background = element_blank()) +
  geom_text_repel(
    data=filter(highlight, Gene.Names.First == 'DAP'|phosphosite %in% c('Q96B36_S183', 'Q96B36_T233', 'Q96B36_T246')),
    aes(label=paste(sub('_HUMAN', '', Entry.Name), gsub('.*_', '', phosphosite))),
    fontface = "bold",
    max.overlaps = 50, size=5, min.segment.length = 0, colour='grey20', force = 20, nudge_x = 1) -> p

print(p)
```



```
ggsave(plot=p, here('results/plots/PROTAC_phospho_limma_results_mTORC1_substrates.png'))
```

```
## Saving 10 x 6 in image
```

```
ggsave(plot=p, here('results/plots/PROTAC_phospho_limma_results_mTORC1_substrates.pdf'))
```

```
## Saving 10 x 6 in image
```

```
crosstab_poi_sig <- phospho_limma_results %>%
  group_by(is_poi=ifelse(phosphosite %in% highlight_val$phosphosite, 'mTORC1 substrate', 'Other'),
    significant=adj.P.Val<0.1) %>%
  tally() %>%
  pivot_wider(names_from = significant, values_from=n, values_fill=0) %>%
  tibble::column_to_rowname('is_poi')

crosstab_poi_sig %>%
  mutate(frac_sig=`TRUE`/(`TRUE`+`FALSE`))
```

```
##           FALSE TRUE  frac_sig
## Other      18943 1124 0.05601236
## mTORC1 substrate    26    3 0.10344828
```

```
print(crosstab_poi_sig)
```

```
##           FALSE TRUE
## Other      18943 1124
## mTORC1 substrate    26    3
```

```
fisher.test(crosstab_poi_sig)
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  crosstab_poi_sig
## p-value = 0.2205
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.3761309 6.3605363
## sample estimates:
## odds ratio
##  1.944541
```

```
crosstab_poi_sig <- phospho_limma_results %>%
  group_by(is_poi=ifelse(phosphosite %in% highlight$phosphosite, 'mTORC1 substrate', 'Other'),
    significant=adj.P.Val<0.1) %>%
  tally() %>%
  pivot_wider(names_from = significant, values_from=n, values_fill=0) %>%
  tibble::column_to_rownames('is_poi')

crosstab_poi_sig %>%
  mutate(frac_sig=`TRUE`/(`TRUE`+`FALSE`))
```

```
##                FALSE TRUE  frac_sig
## Other                18744 1113 0.05605076
## mTORC1 substrate      225    14 0.05857741
```

```
print(crosstab_poi_sig)
```

```
##                FALSE TRUE
## Other                18744 1113
## mTORC1 substrate      225    14
```

```
fisher.test(crosstab_poi_sig)
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  crosstab_poi_sig
## p-value = 0.7783
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.5618574 1.8030059
## sample estimates:
## odds ratio
##  1.047898
```

6 Save results

```
phospho_limma_results %>%
  select(UniprotID, phosphosite, phospho_pos, logFC, P.Value, adj.P.Val,
         Protein.names=Protein.names.y, Gene.Names, Gene.Names.First, site_seq) %>%
  write.table(here('results/PROTAC_phospho_limma_results.tsv'),
             sep='\t', row.names=FALSE, quote=FALSE)
```

Finally, session info:

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.6.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_GB.UTF-8/en_GB.UTF-8/en_GB.UTF-8/C/en_GB.UTF-8/en_GB.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats4 stats graphics grDevices utils datasets methods
## [8] base
##
## other attached packages:
## [1] ggbeeswarm_0.7.3 tidyr_1.3.2
## [3] dplyr_1.2.1 ggrepel_0.9.7
## [5] ggplot2_4.0.3 limma_3.66.0
## [7] here_1.0.2 biomasslmb_0.0.5
## [9] QFeatures_1.20.0 MultiAssayExperiment_1.36.2
## [11] SummarizedExperiment_1.40.0 Biobase_2.70.0
## [13] GenomicRanges_1.62.1 Seqinfo_1.0.0
## [15] IRanges_2.44.0 S4Vectors_0.48.1
## [17] BiocGenerics_0.56.0 generics_0.1.4
## [19] MatrixGenerics_1.22.0 matrixStats_1.5.0
##
## loaded via a namespace (and not attached):
## [1] DBI_1.3.0 rlang_1.2.0 magrittr_2.0.5
## [4] clue_0.3-68 otel_0.2.0 compiler_4.5.2
## [7] RSQlite_3.53.2 png_0.1-9 systemfonts_1.3.1
## [10] vctr_0.7.3 reshape2_1.4.5 stringr_1.6.0
## [13] ProtGenerics_1.42.0 pkgconfig_2.0.3 crayon_1.5.3
## [16] fastmap_1.2.0 backports_1.5.0 XVector_0.50.0
## [19] labeling_0.4.3 rmarkdown_2.31 visdat_0.6.0
## [22] ragg_1.5.0 purrr_1.2.2 bit_4.6.0
## [25] xfun_0.59 cachem_1.1.0 blob_1.3.0
```

## [28] DelayedArray_0.36.1	cluster_2.1.8.2	R6_2.6.1
## [31] stringi_1.8.7	RColorBrewer_1.1-3	genefilter_1.92.0
## [34] Rcpp_1.1.1-1.1	knitr_1.51	BiocBaseUtils_1.12.0
## [37] Matrix_1.7-4	splines_4.5.2	igraph_2.3.2
## [40] tidyselect_1.2.1	abind_1.4-8	yaml_2.3.12
## [43] lattice_0.22-9	tibble_3.3.1	plyr_1.8.9
## [46] withr_3.0.3	KEGGREST_1.50.0	S7_0.2.2
## [49] evaluate_1.0.5	uniprotREST_1.0.0	survival_3.8-6
## [52] zip_3.0.0	Biostrings_2.78.0	pillar_1.11.1
## [55] corrplot_0.95	checkmate_2.3.4	rprojroot_2.1.1
## [58] scales_1.4.0	xtable_1.8-8	glue_1.8.1
## [61] lazyeval_0.2.3	tools_4.5.2	robustbase_0.99-7
## [64] openxlsx_4.2.8.1	annotate_1.88.0	XML_3.99-0.23
## [67] grid_4.5.2	MsCoreUtils_1.22.1	AnnotationDbi_1.72.0
## [70] beeswarm_0.4.0	vipor_0.4.7	naniar_1.1.0
## [73] cli_3.6.6	textshaping_1.0.4	S4Arrays_1.10.1
## [76] AnnotationFilter_1.34.0	gtable_0.3.6	DEoptimR_1.2-0
## [79] digest_0.6.39	SparseArray_1.10.10	farver_2.1.2
## [82] memoise_2.0.1	htmltools_0.5.9	lifecycle_1.0.5
## [85] httr_1.4.8	statmod_1.5.1	bit64_4.8.2
## [88] MASS_7.3-65		